

Sql Notes5

Temporary Table:

A temporary table is a table that exists only for the duration of a session, transaction or connection (depending on the database system). It's commonly used for intermediate results, staging data or simplifying complex queries.

```
CREATE TEMPORARY TABLE temp_employees (  
id INT,  
name VARCHAR(100)  
);
```

```
INSERT INTO temp_employees  
VALUES (1, 'John');
```

```
SELECT * FROM temp_employees;
```

```
DROP TABLE temp_employees;
```

Useful for:

- Breaking complex queries into steps
- ETL/data transformation
- Reporting
- Performance optimization

Views:

A View is a virtual table based on the result of a query. It does not usually store data itself (except materialized views in some DBs). Instead, it stores the SQL query definition.

```
CREATE VIEW employee_view AS  
SELECT id, name, department  
FROM employees  
WHERE active = 1;
```

Stored Procedures:

A **Stored Procedure** is a saved collection of SQL statements stored inside the database that can be executed whenever needed.

It helps with:

- reusable business logic
- performance optimization
- security
- reducing repeated SQL code

DELIMITER //

```
CREATE PROCEDURE GetEmployees()  
BEGIN  
SELECT * FROM employees;  
END //
```

DELIMITER ;

Delimiter:

Temporarily changes statement ending symbol.

Call:

Executes stored procedure.

1. IN Parameter

Used to pass values into the procedure.

DELIMITER //

```
CREATE PROCEDURE GetEmployeeByDept(  
    IN dept_name VARCHAR(50)  
)
```

```
BEGIN

  SELECT *

  FROM employees

  WHERE department = dept_name;

END //
```

2. OUT Parameter

Used to return a value from the procedure.

```
DELIMITER //
```

```
CREATE PROCEDURE GetEmployeeCount(

  OUT total INT

)

BEGIN

  SELECT COUNT(*) INTO total

  FROM employees;

END //
```

3. INOUT Parameter

Acts as both input and output.

```
DELIMITER //
```

```
CREATE PROCEDURE IncreaseValue(

  INOUT num INT

)
```

BEGIN

SET num = num + 10;

END //

DELIMITER ;

Advantages of Stored Procedures:

1. Reusability

Write once, use many times.

2. Better Performance

Execution plans may be cached.

3. Security

Users can execute procedures without direct table access.

4. Reduced Network Traffic

One call can execute multiple SQL statements.

Disadvantages:

- Harder debugging
- DB-specific syntax
- Complex logic can become difficult to maintain

Dynamic SQL in Stored Procedures:

Dynamic SQL means creating and executing SQL statements at runtime instead of writing fixed SQL queries.

Useful when:

- table names change dynamically

- column names are dynamic
- optional filters are needed
- query structure changes based on input